

Lesson 26: Polynomial Inequalities

Criteria	Excellent (4)	Proficient (3)	Developing (2)	Beginning (1)
Setup & Form	Correctly rearranges inequality to $P(x) \leq 0$ or $P(x) > 0$ and accurately factors the polynomial.	Rearranges inequality correctly but makes minor errors in factoring.	Incorrectly rearranges or struggles to factor the polynomial.	Fails to set the inequality to zero or cannot factor.
Critical Points (Zeros)	Correctly identifies all real roots (zeros) of the polynomial, including multiplicity.	Identifies most zeros, but misses one or includes a non-real root.	Struggles to find zeros; uses incorrect factoring methods.	Cannot identify critical points.
Sign Chart / Interval Analysis	Accurately tests all intervals using correct test points and determines sign of each region.	Correct testing method, but 1-2 errors in calculating signs for regions.	Incorrect test points selected or improper interpretation of sign changes.	No sign chart or test points provided.

Interval Notation Solution	Expresses the final solution set perfectly in interval notation, using correct brackets or parentheses	Writes interval notation, but confuses brackets and parentheses (e.g., includes endpoints when they should be excluded).	Uses proper notation format, but intervals do not match sign chart.	Incorrect notation, or does not use interval notation.
Accuracy	100% correct solution set with no mistakes in steps.	Minor computation error, but steps and method are correct.	Significant errors in logic or computation.	Fails to find a valid solution.